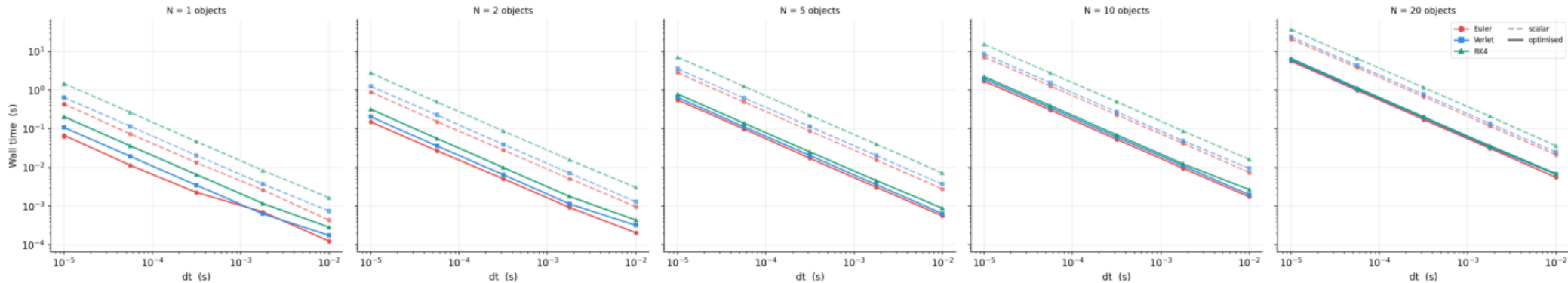
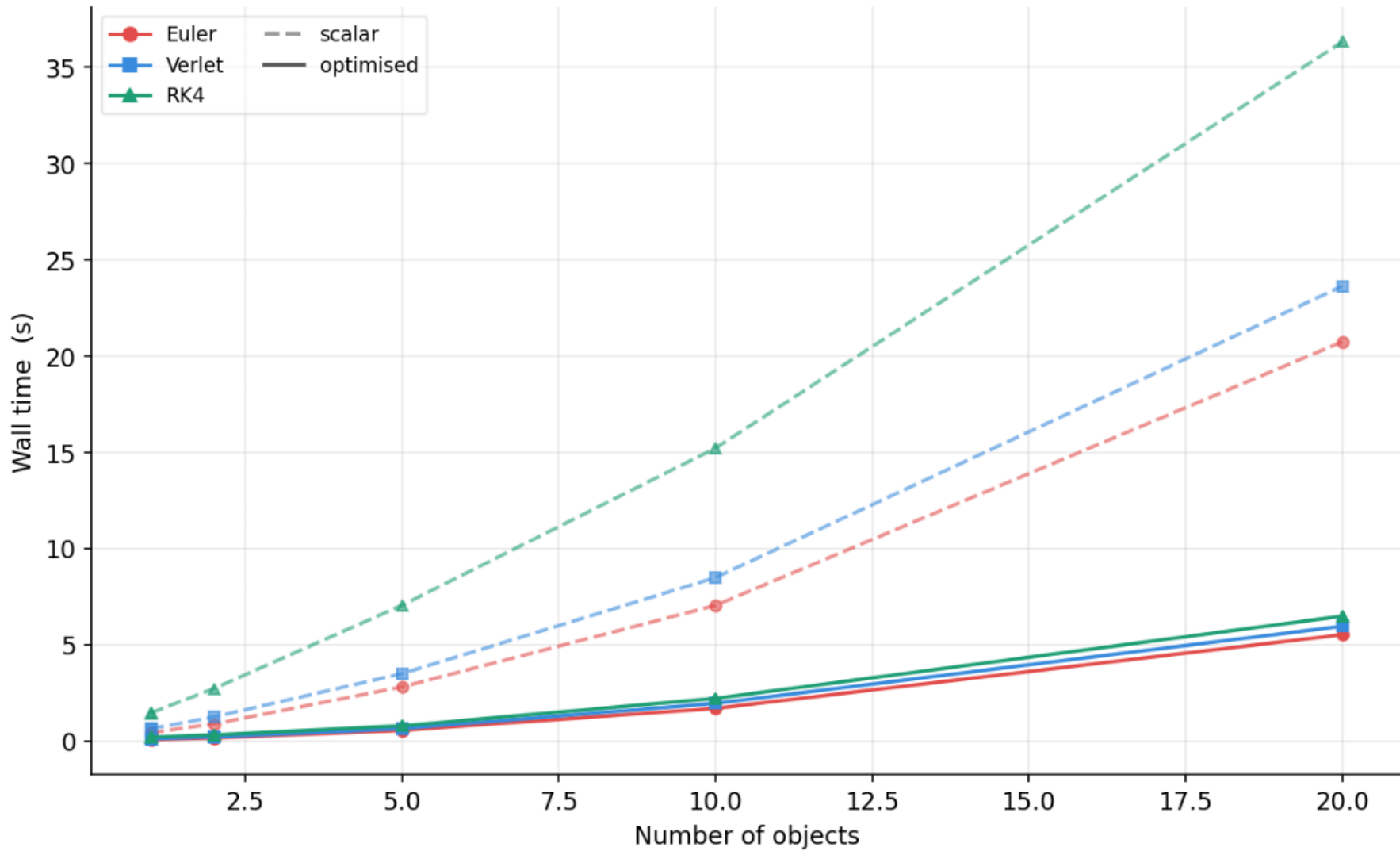


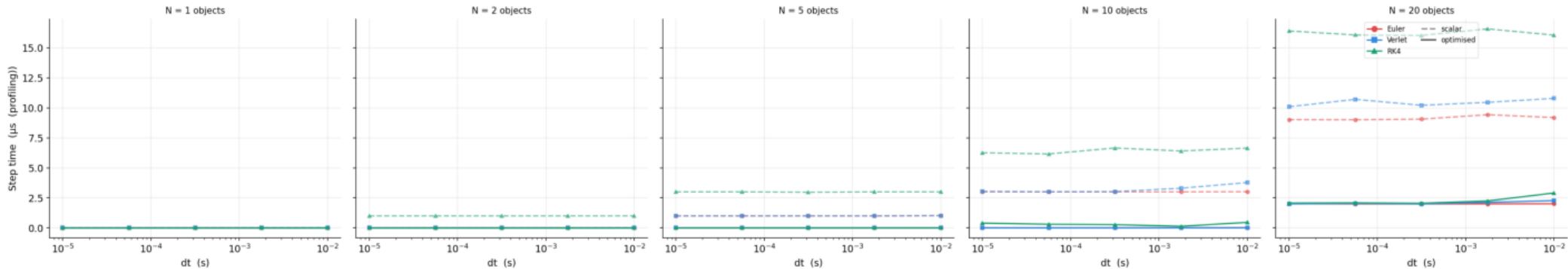
Wall time vs dt — $O(1/dt)$ scaling expected



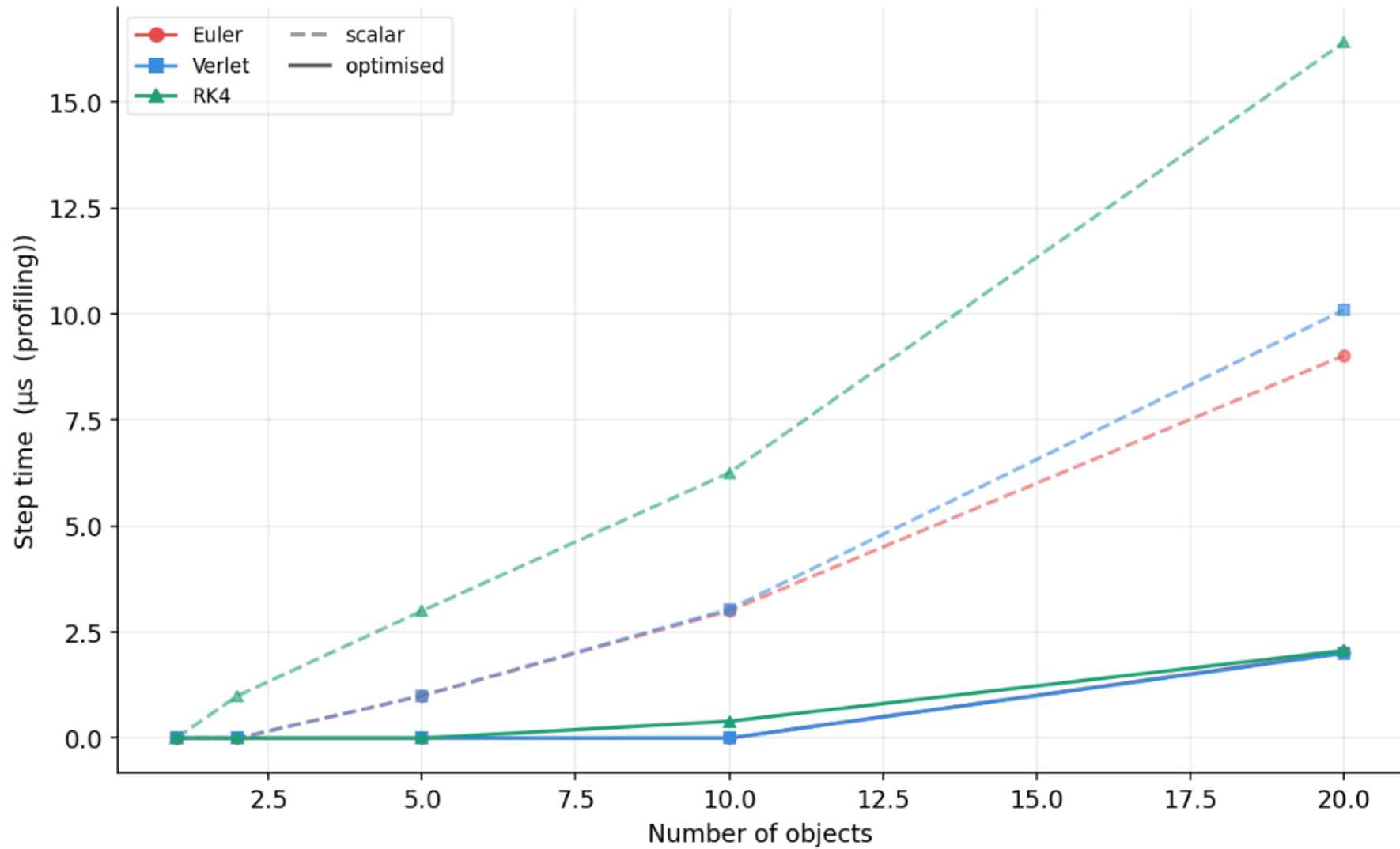
Wall-time scaling with object count — $O(N^2)$ collision expected
($dt = 1.00e-05$ s)



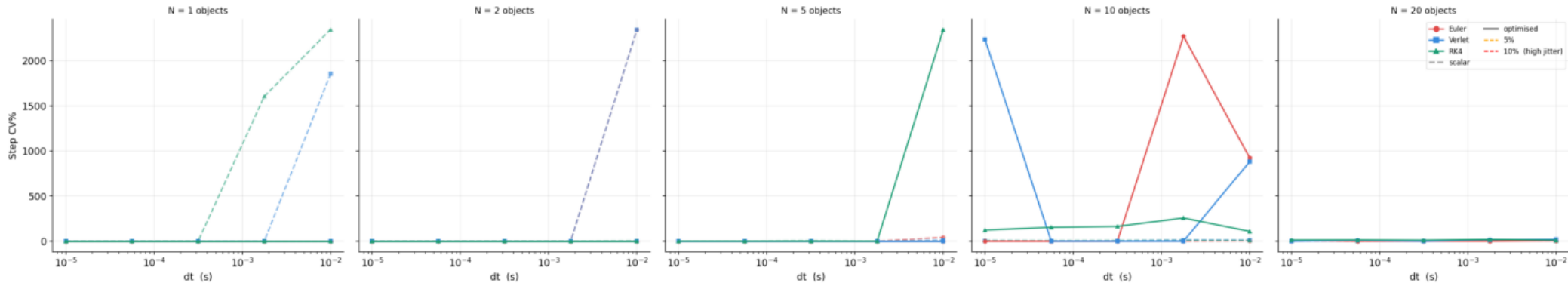
Step time vs dt
(plateau at fine dt = true per-step cost)



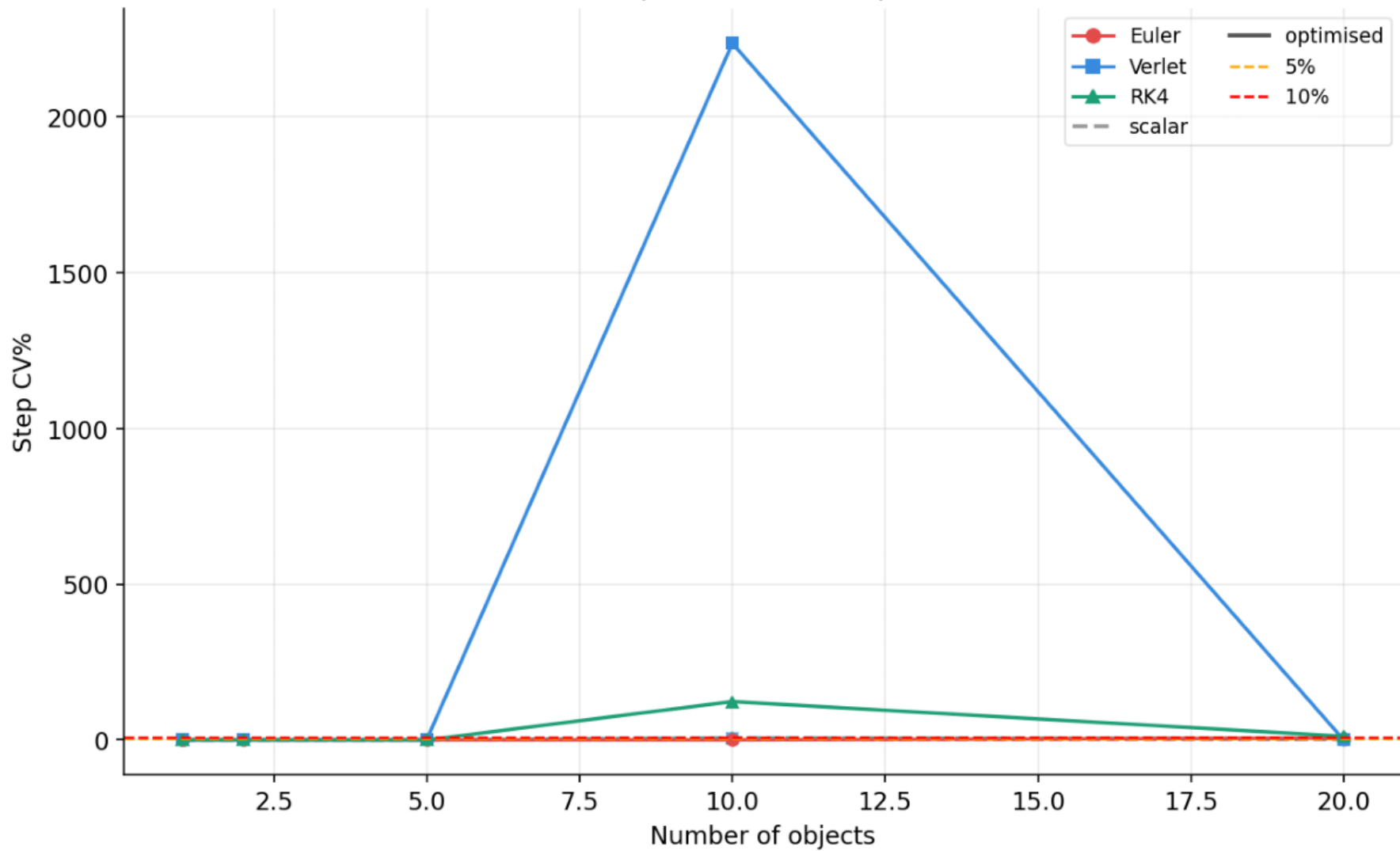
Step-time scaling with object count
(dt = 1.00e-05 s)



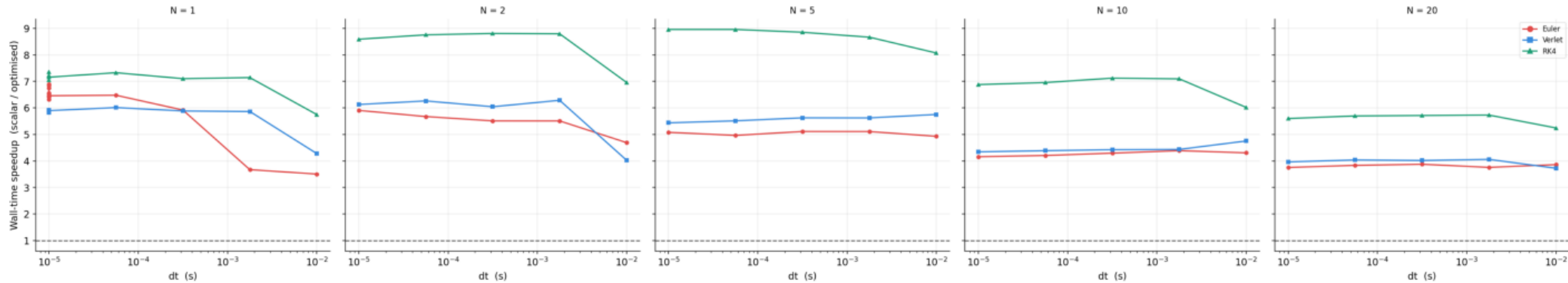
Per-step jitter vs dt (low CV → regular cost)



Per-step jitter vs object count
(dt = 1.00e-05 s)

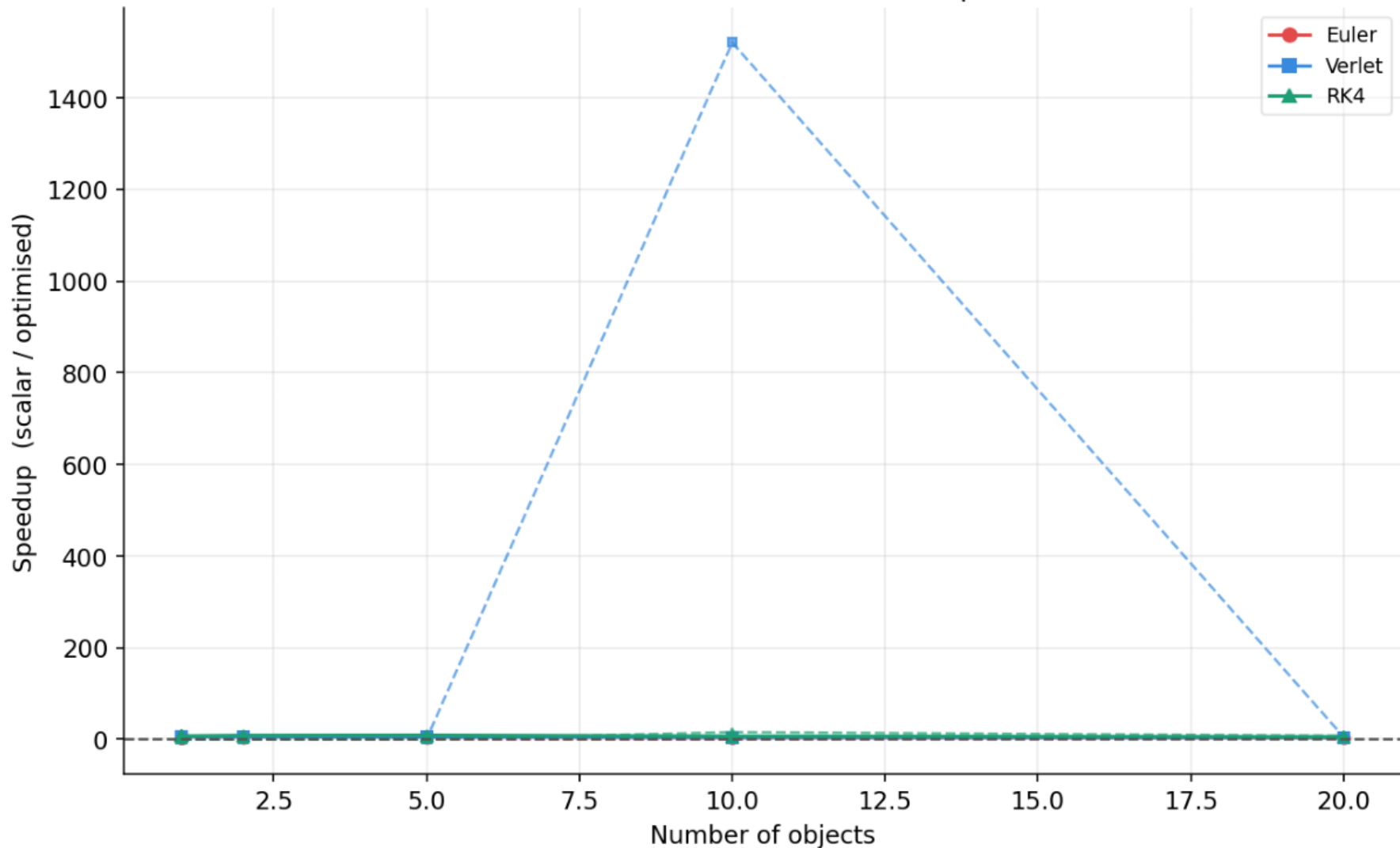


Scalar → Optimised wall-time speedup vs dt (> 1 = optimised wins)



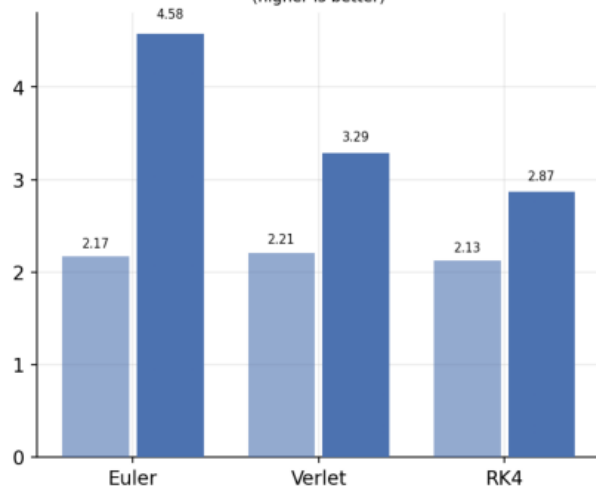
Speedup vs object count (dt = 1.00e-05 s)

solid = wall-time · dashed = step-time

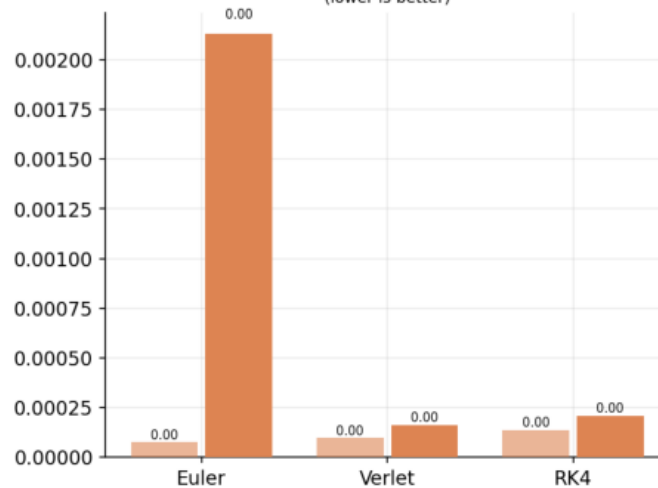


Hardware Counters — perf stat (grouped by solver, shading = profile)

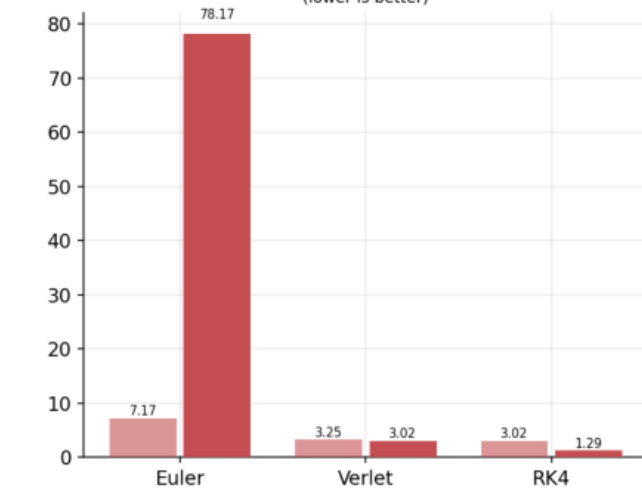
IPC
(higher is better)



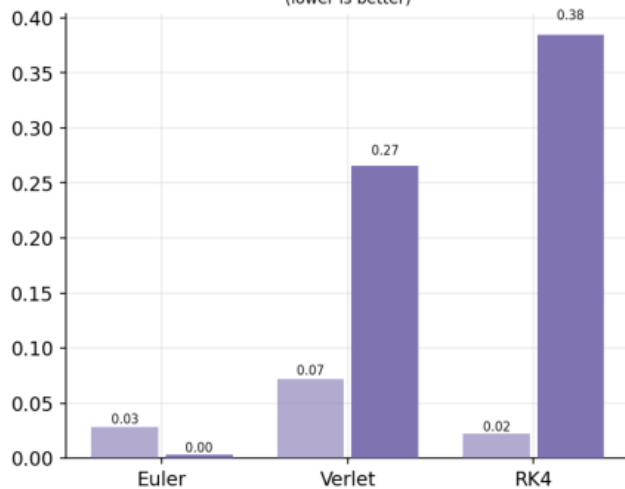
L1 miss %
(lower is better)



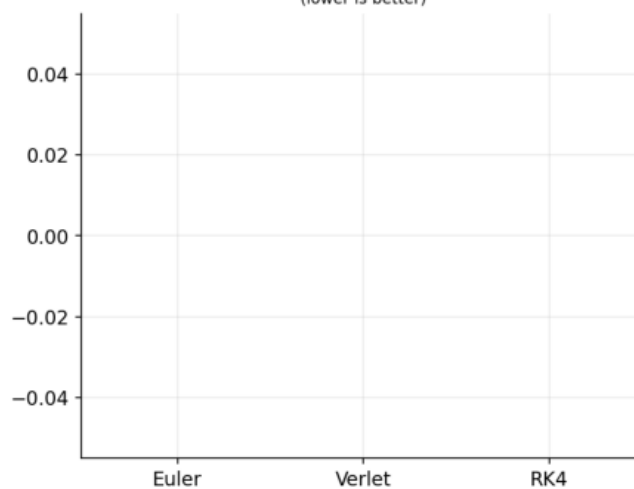
LLC miss %
(lower is better)



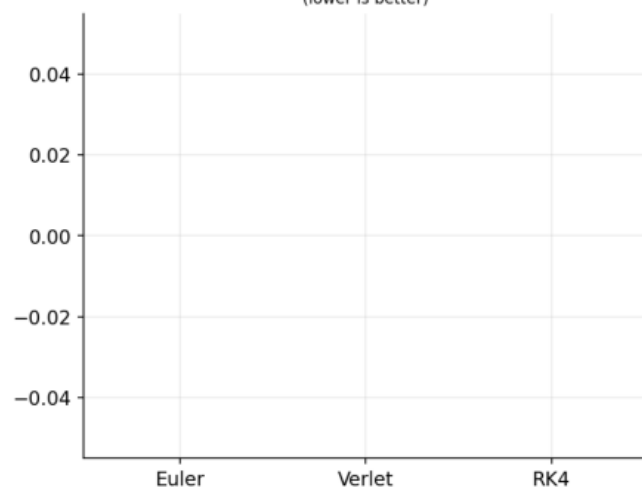
branch miss %
(lower is better)



stall fe %
(lower is better)

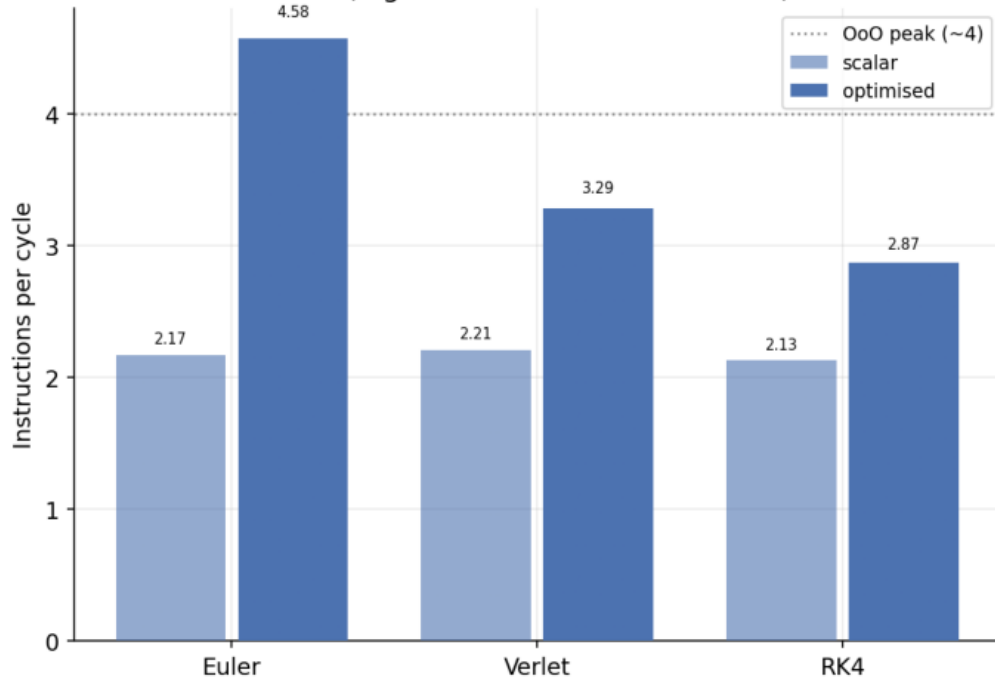


stall be %
(lower is better)

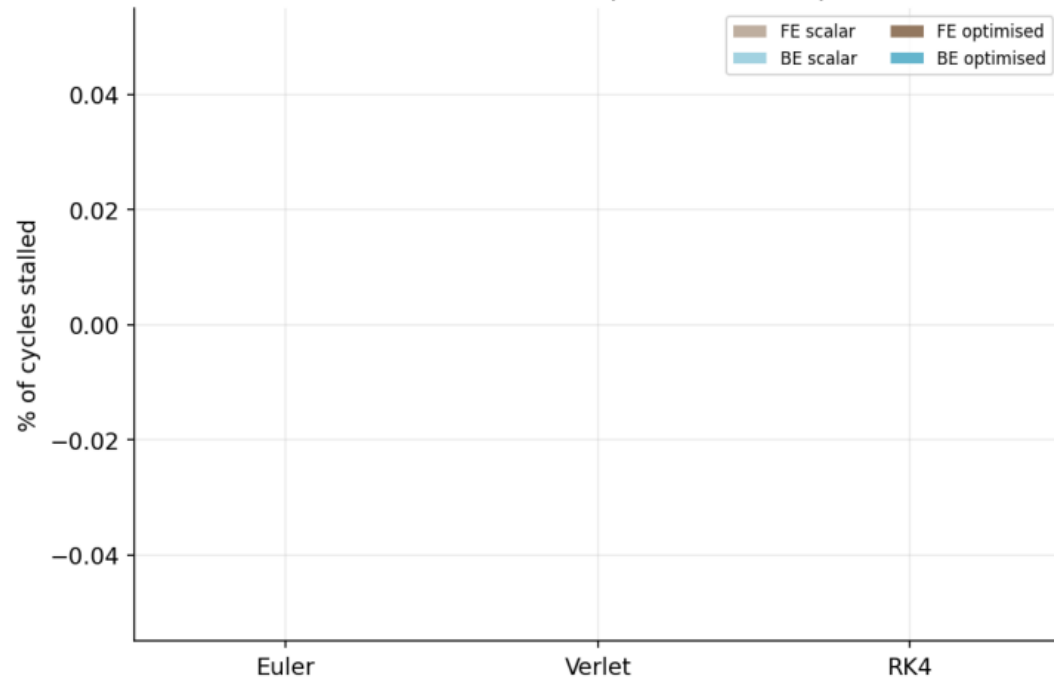


-- scalar — optimised

IPC (higher = better CPU utilisation)

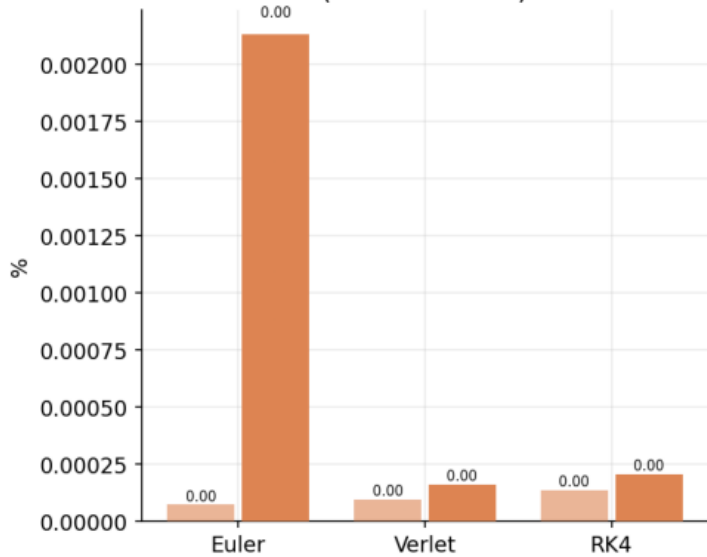


Stall breakdown (lower = better)

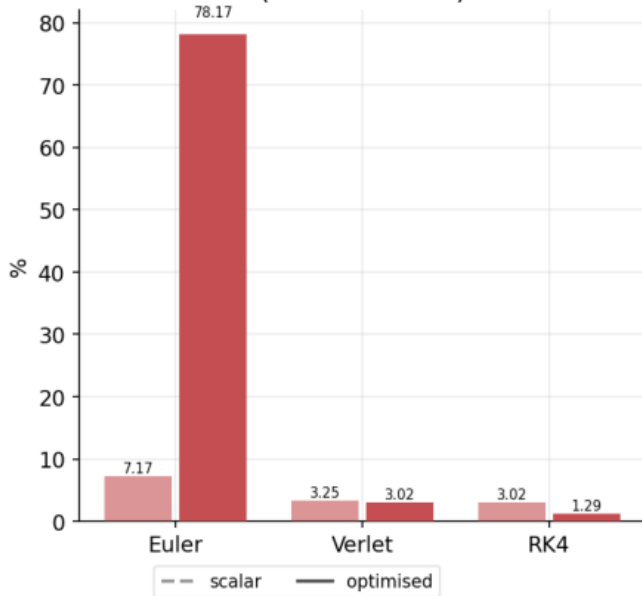


Cache & Branch Performance — perf stat

L1 D-cache miss rate (%)
(lower is better)



LLC miss rate (%)
(lower is better)



Branch miss rate (%)
(lower is better)

